The POSIX Lexer Cubic Bound: the Set-Ledger Gate (Open Problem)

POSIX backref/cubic project — secretary session
repository `github.com/hellotommmy/posix`, branch `codex/backref-values`

2026-06-12

Abstract

POSIX lexing by Brzozowski/Antimirov derivatives stays fast only if the simplified derivative states stay small. This note gives a fully self-contained statement of the one inequality this project still has to prove: after one step of the strongly simplified Antimirov-style row transition, the total size of the *deduplicated* set of opened linear forms is cubic in the size of the original regular expression. All definitions below are faithful translations of the checked Isabelle/HOL definitions in the repository (exact names in the table of Section 8); every fact labelled "checked" is proved in Isabelle with no `sorry`. Section 5 records what is already proved, Section 6 what is *refuted* and must not be attempted again.

1 Regular expressions and size

Regular expressions over an alphabet Σ (Isabelle skeleton type rrexp, restricted to its classical—"legacy"—fragment[1]):

$$r \ ::= \ \mathbf{0} \ \mid \ \mathbf{1} \ \mid \ c \ \mid \ r_1 \cdot r_2 \ \mid \ \Sigma[r_1, \ldots, r_n] \ \mid \ r^* \ \mid \ r^{\{n\}} \qquad (c \in \Sigma, \ n \in \mathbb{N})$$

where $\Sigma[\,\cdot\,]$ is a *list* alternative (ordered, possibly with duplicates) and $r^{\{n\}}$ is bounded repetition (RNTIMES). Size $|\cdot|$ and list size $\|\cdot\|$ (rsize, rsizes):

$$|\mathbf{0}| = |\mathbf{1}| = |c| = 1, \quad |r_1 \cdot r_2| = 1 + |r_1| + |r_2|, \quad |\Sigma rs| = 1 + \textstyle\sum_{r \in rs} |r|,$$

$$|r^*| = 1 + |r|, \qquad |r^{\{n\}}| = 1 + |r| + n, \qquad \|rs\| = \textstyle\sum_{r \in rs} |r| \quad (rs \text{ a list}).$$

Note the repetition counter n itself counts toward size. Nullability ν (rnullable) is standard; $\nu(r^{\{0\}})$ holds.

2 The transition pipeline

List helpers. flts (rflts) deletes $\mathbf{0}$ members and splices one level of nested alternatives: $\mathsf{flts}(\mathbf{0}\#rs) = \mathsf{flts}\,rs$; $\mathsf{flts}((\Sigma ts)\#rs) = ts \mathbin{+\!\!+} \mathsf{flts}\,rs$; otherwise the head is kept. nub (rdistinct, with empty accumulator) keeps the first occurrence of each member, preserving order. The smart alternative constructor $\oplus$ (rsimp_ALTs) maps $[\,] \mapsto \mathbf{0}$, $[r] \mapsto r$, and longer lists $rs \mapsto \Sigma rs$.

[1] The formal skeleton carries three additional proof-only backreference constructors (RBACKREF4, RHALF, RRESIDUE). They have no partial derivatives and are excluded here by the predicate legacy_rrexp; the open problem lives entirely in the classical fragment.

Sequence normalizer. σ ($\mathsf{rsimp4\_SEQ\_atom}$) builds a right-associated sequence while erasing units:

$$\sigma(\mathbf{0},t)=\mathbf{0},\quad \sigma(\mathbf{1},t)=t,\quad \sigma(p\cdot q,\,t)=\sigma(p,\sigma(q,t)),$$

and for an atom a (none of $\mathbf{0},\mathbf{1},\cdot$): $\sigma(a,\mathbf{0})=\mathbf{0}$, $\sigma(a,\mathbf{1})=a$, otherwise $\sigma(a,t)=a\cdot t$. On top of it, $\sigma^\star$ ($\mathsf{rsimp7\_SEQ\_atom}$) absorbs adjacent equal stars:

$$\sigma^\star(p^*,p^*)=p^*,\qquad \sigma^\star(p^*,\,p^*\cdot k)=p^*\cdot k,\qquad \sigma^\star(p,t)=\sigma(p,t)\text{ otherwise.}$$

Partial derivatives. The Antimirov partial-derivative list pder_c ($\mathsf{rpder\_list}$):

$$\begin{aligned}
\mathsf{pder}_c(\mathbf{0})=\mathsf{pder}_c(\mathbf{1})&=[\,],\qquad \mathsf{pder}_c(d)=\text{if } c{=}d \text{ then } [\mathbf{1}] \text{ else } [\,],\\
\mathsf{pder}_c(\Sigma rs)&=\mathsf{concat}(\mathsf{map}\ \mathsf{pder}_c\ rs),\\
\mathsf{pder}_c(p\cdot q)&=\mathsf{map}\,(\lambda x.\,\sigma(x,q))\,(\mathsf{pder}_c\,p) \mathbin{+\!\!+} (\text{if } \nu(p) \text{ then } \mathsf{pder}_c\,q \text{ else } [\,]),\\
\mathsf{pder}_c(p^*)&=\mathsf{map}\,(\lambda x.\,\sigma(x,p^*))\,(\mathsf{pder}_c\,p),\\
\mathsf{pder}_c(p^{\{n\}})&=\text{if } n{=}0 \text{ then } [\,] \text{ else } \mathsf{map}\,(\lambda x.\,\sigma(x,p^{\{n-1\}}))\,(\mathsf{pder}_c\,p),
\end{aligned}$$

and its normalized form $\mathsf{npder}_c(r)=\mathsf{map}\,(\lambda x.\,\sigma(x,\mathbf{1}))\,(\mathsf{pder}_c\,r)$ ($\mathsf{rpder\_norm\_list}$).

Multi-step factored states. One factored step on a row list ($\mathsf{afactored\_step} = \mathsf{rpder\_norm\_rows}$) and its iteration ($\mathsf{afactored1}$):

$$\mathsf{step}_c(rs)=\mathsf{nub}\big(\mathsf{flts}\big(\mathsf{concat}(\mathsf{map}\ \mathsf{npder}_c\ rs)\big)\big),\qquad F(r,\varepsilon)=[r],\quad F(r,\,s\,c)=\mathsf{step}_c(F(r,s)).$$

Note: *no* strong pruning happens inside F; the strong simplifier enters only in the single step that the open problem measures.

Strong simplifier. The pairwise shared-suffix pruner ($\mathsf{rsimpStrong\_prune\_pair\_raw}$): an earlier row e prunes a later row l by

$$P(e,l)=\begin{cases}\sigma^\star\big(\oplus(\,ms\setminus ls\,),\,k\big) & \text{if } e=(\Sigma ls)\cdot k \text{ and } l=(\Sigma ms)\cdot k,\\ l & \text{otherwise,}\end{cases}$$

where $ms\setminus ls$ removes from the list ms every member occurring in ls ($\mathsf{rprune\_eq\_against}$), keeping order. The scan ($\mathsf{rsimpStrong\_prune\_rows\_raw}$) folds P over the already emitted rows:

$$\mathsf{scan}(seen,[\,])=[\,],\qquad \mathsf{scan}(seen,\,r\#rs)=r'\#\,\mathsf{scan}(r'\#seen,\ rs),$$

where $r'=\big(\mathrm{fold}_{e\in seen}P(e,\cdot)\big)(r)$, and $\mathsf{prune}(rs)=\mathsf{scan}([\,],rs)$. The strong simplifier S ($\mathsf{rsimpStrong\_raw}$):

$$\begin{aligned}
S(\mathbf{0})=\mathbf{0},\quad S(\mathbf{1})&=\mathbf{1},\quad S(c)=c,\qquad S(p\cdot q)=\sigma^\star(S(p),S(q)),\\
S(\Sigma rs)&=\oplus\big(\mathsf{nub}\big(\mathsf{flts}\big(\mathsf{prune}\big(\mathsf{flts}(\mathsf{map}\ S\ rs)\big)\big)\big)\big),\qquad S(p^{\{n\}})=S(p)^{\{n\}},\\
S(p^*)&=\begin{cases}\mathbf{1} & \text{if } S(p)\in\{\mathbf{0},\mathbf{1}\},\\ q^* & \text{if } S(p)=q^*,\\ S(p)^* & \text{otherwise.}\end{cases}
\end{aligned}$$

The strong one-step row transition. ($\mathsf{rpder\_strong\_rows\_raw}$; this is the object the problem measures.)

$$\Delta_c(rs)\ =\ \mathsf{nub}\Big(\mathsf{flts}\Big(\mathsf{prune}\Big(\mathsf{flts}\big(\mathsf{concat}\,(\mathsf{map}\ (\mathsf{map}\ S\circ\mathsf{npder}_c)\ rs)\big)\Big)\Big)\Big).$$

We also name the *generated* rows before simplification: $\mathsf{gen}_c(rs)=\mathsf{concat}(\mathsf{map}\ \mathsf{npder}_c\ rs)$.

3 The ledger objects

The *opener* D (row_dlforms) unfolds a row into its deduplicated linear forms[2]:

$$D(\mathbf{0}) = \emptyset, \qquad D(\Sigma rs) = \bigcup_{q\in rs} D(q), \qquad D\big((\Sigma ps)\cdot k\big) = \bigcup_{p\in ps} D\big(\sigma^\star(p,k)\big),$$

and $D(q) = \{q\}$ for every other shape. For a row list, $D(rs) = \bigcup_{q\in rs} D(q)$ (row_dlformss). The *set ledger* of a finite set U of regexes (rsize_set):

$$\|U\| \;=\; \sum_{q\in U} |q|.$$

4 The open problem

Conjecture 1 (set-ledger cubic gate)**.** *For every classical (legacy) regular expression r, every string $s \in \Sigma^*$ and every character $c \in \Sigma$:*

$$\boxed{\big\|\, D\big(\Delta_c(F(r,s))\big) \,\big\| \;\le\; 2\,\big(|r|+3\big)^3.}$$

In words: run the normalized Antimirov row automaton from r over s, apply *one* strongly simplified step for c, open every resulting row into linear forms, deduplicate *across the whole union*, and sum the sizes of the distinct opened rows. The claim is that this total is cubic in the size of the original expression — uniformly in s, whose length does not appear on the right-hand side.

Remark 1. The constant $2(\cdot+3)^3$ is the project's working target; a checked bound $K(|r|+3)^3$ for any absolute constant K, or for the $r^{\{n\}}$-free sub-fragment, is already valuable (see Section 7).

Remark 2. The inequality is stated for arbitrary r in the consuming contracts; the checked supporting facts that feed it (head linearity, the discharged gate instance below) carry the hypothesis legacy_rrexp r, so Conjecture 1 should be read with that hypothesis. Empirically the left-hand side grows only *quadratically* on exhaustive and random probes (checkpoint 2026-06-12, "multi-step union ledger probe").

Remark 3 (why this matters). The conjecture is exactly the missing *size half* of the per-step state budget for the strongly simplified POSIX lexer. The repository already contains checked contracts (rpder_strong_dcanon_rows_raw_afactored1_actual_dlforms_tight_cubic_contractI, ..._budgetsI) that take Conjecture 1 as their only numeric assumption and return the full canonical-row contract: language correctness of the canonical rows together with cubic bounds on their length, cardinality, linear-term count, and total size. The value/POSIX-correctness side of the lexer is independently checked and waiting (strong_deferred_memo_lexer_eq_lexer etc.).

5 What is already checked

All of the following are Isabelle-checked, unconditional unless stated.

1. *Per-row bounds.* $\#D(q) \le |q|$ (card_row_dlforms_le_rsize)
   $\|D(q)\| \le |q|^2$ (rsize_set_row_dlforms_le_rsize_sq)

2. *Union bridges.* $\|D(rs)\| \le \sum_{q\in rs} |q|^2$, instantiated to the actual one-step output (rsize_set_row_dlformss_..._le_sum_rsize_sq)

[2] Recursion terminates by the size measure: when $p \in ps$, $|\sigma^\star(p,k)| < |(\Sigma ps)\cdot k|$ (rsize_rsimp7_SEQ_atom_member_lt_RSEQ_RALTS).

3. *Card half, one degree.* $\#D(\Delta_c(rs)) \le \|\mathsf{gen}_c(rs)\|$ (card_row_dlformss_rpder_strong_rows_raw_le_generated)

4. *Deleter chain.* $\|\Delta_c(rs)\| \le \|\mathsf{gen}_c(rs)\|$, and the same budget bounds length, cardinality and linear-term count of $\Delta_c(rs)$ (rsizes_rpder_strong_rows_raw_le, rpder_strong_rows_raw_generated_budget)

5. *Head linearity (legacy).* every sequence head h with $h \cdot t \in D(\Delta_c(F(r,s)))$ satisfies $|h| \le 1 + 2|r|$ (actual_union_seq_head_size_linear)

6. *Discharged gate instance (legacy; 2026-06-12).* With $H = 1 + 2|r|$ and $M = 1 + H + \|\mathsf{gen}_c(F(r,s))\|$:

$$\big\|D\big(\Delta_c(F(r,s))\big)\big\| \;\le\; \underbrace{\big\|\mathsf{nonseq}(U)\big\|}_{(1)} + \underbrace{\#\mathsf{fronts}(r,sc)\cdot \textstyle\sum_{t\in\mathsf{tails}(U)}\big(1{+}H{+}|t|\big)}_{(2)}$$
$$+ \underbrace{\mathsf{pairbudget}(r,s,c)\cdot M}_{(3)}$$

where $U = D(\Delta_c(F(r,s)))$, $\mathsf{nonseq}(U)$ are its non-sequence members, $\mathsf{tails}(U)$ the set of tails of its sequence members, $\mathsf{fronts}(r,u)$ a syntactic carrier of front terms (strong_derivative_front_terms, members linear by item 5), and pairbudget the shared-suffix pair budget of the one-step dlform universe (raw_shared_prune_active_suffix_pair_budget over afactored1_strong_dlform_universe)

(actual_union_pair_budget_gate_instance)

7. *Summand (1) is cubic.* (rsize_set_rnonseq_members_. . . _afactored1_cubic)

8. *Fragment result.* If r is $r^{\{n\}}$-free (and normal-form), then $\|F(r,s)\| \le 2(2|r|+3)^3$ (rsizes_afactored1_rntimes_free_rsize_cubic)

9. *Summand (3) vanishes (2026-06-12, 20:44).* The dlform universe contains only opened rows, and opened sequence heads are never alternatives (item 6's carrier), so no universe member has the keyed shape $(\Sigma\, rows)\cdot k$; hence the active-suffix key set is empty and $\mathsf{pairbudget}(r,s,c) = 0$ identically. The gate instance has exactly two summands. (pair_budget_afactored1_strong_dlform_universe_zero, actual_union_gate_two_summands)

Current reduction (as of 2026-06-12, commit `cfe3636`). By items 6–9 the conjecture reduces to a *single* numeric obligation: summand (2), the tail-weight side, must be cubic. The exact decomposition on record is

$$\|U\| \;=\; \underbrace{\|\mathsf{nonseq}(U)\|}_{\text{cubic, checked}} \;+\; \underbrace{\#\{\text{sequence members}\}\cdot O(|r|)}_{\text{heads linear, checked}} \;+\; \sum_{t\in\mathsf{tails}(U)} \mathsf{bucket}(t)\cdot|t|,$$

where $\mathsf{bucket}(t)$ counts the sequence members sharing tail t. The two load-bearing quantities are $\#U$ (empirically linear; best checked bound one degree in $\|\mathsf{gen}\|$) and $\mathsf{bucket}(t)$ (empirically linear; best checked bound cubic). Any other route to Conjecture 1 is equally acceptable, but must not pass through the refuted quantities below.

6 What is refuted (do not attempt)

Theorem 1 (duplicated list cost is not polynomial; checked)**.** *Define the* list *opener* D^{list} *by the same recursion as* D *with concatenation instead of union (no deduplication), and the pipeline list cost*

$$\mathsf{listcost}(r,s,c) = \sum_{p\,\in\,\mathsf{gen}_c(F(r,s))} \big\|D^{\mathsf{list}}\big(S(p)\big)\big\|.$$

Let $\pi = (\mathbf{1}+a)\cdot(\mathbf{1}+b)$ *and the tower* $T_0 = k\cdot l$, $T_{n+1} = (\mathbf{1}+\pi)\cdot T_n$ *(here* $\mathbf{1}+x$ *abbreviates* $\Sigma[\mathbf{1},x]$*;* a,b,k,l *characters). Then (all checked):*

$$|T_n| = 10n+3, \qquad \mathsf{len}\, D^{\mathsf{list}}(T_n) = 3\cdot 2^n - 2, \qquad S(T_n) = T_n,$$

and behind one guard character g, $\mathsf{listcost}(g\cdot T_{24},\, \varepsilon,\, g) \geq 3\cdot 2^{24} - 2 = 50{,}331{,}646 > 2(245+3)^3 = 30{,}505{,}984$*, refuting the cubic budget for* listcost *(afactored1_strong_dlform_list_cost_cubic_false).*

The mechanism: on $(\Sigma[\mathbf{1},\pi])\cdot T$ the list opener recurses into *both* branches; the $\mathbf{1}$ branch collapses to T and reproduces T's entire opened list, the π branch walks it once more — so the length doubles per level, while the regex grows linearly. The tower is a fixpoint of S, so simplification cannot rescue the cost. Crucially, both branches reach the *same* tail rows: the deduplicated union merges what the list multiplies, which is why Conjecture 1 survives this family (quadratically). Consequences and further checked refutations:

1. Any proof route needing a polynomial bound on listcost, D^{list}-length, or any per-row duplicated multiset is dead (Theorem 1).

2. Generic square-sum wrappers ($\sum_q |q|^2 \leq \|rs\|^2$, or len $\times$ max^3 products) are true but lose a degree: they yield quartic, not cubic, bounds.

3. The deep-frontier linear-card premise $\#(\text{deep frontier}) \leq \text{awidth} + |r| + 3$ is *false* in general — $r^{\{n\}}$ multiplies alternation branches without paying into the budget (checked witness with cardinality 49 against budget 39; apder_deep_frontier_linear_card_false); only the $r^{\{n\}}$-free version survives. The front-linear variant is likewise refuted (checked, 2026-06-12).

4. Generic owner-closure counting is exponential (checked abstract counterexample, 2026-06-12).

5. S is *not* pointwise D-cost monotone: there is a checked p with $\|D(S(p))\| = 12 > 10 = \|D(p)\|$ (rsimpStrong_raw_row_dlforms_cost_not_monotone). Do not assume simplification shrinks the ledger row by row.

7 Acceptance criteria

A solution is a proof of Conjecture 1 checked by Isabelle2025-2 in session `Posix` of the repository (branch `codex/backref-values`), with no `sorry`/`oops`/axioms, statement guards passing, building via the repository wrapper (`scripts/codex-isabelle-build-posix.ps1`). Per the 2026-06-12 bounty overlay (simulated USD): 12,000 for the final checked theorem (or a checked equivalent contract) preserving exact POSIX value behaviour; 5,000 for a major checked bridge that reduces it to a clearly smaller, non-false local obligation (the single remaining tail-weight obligation above is the current such target); 3,000 for a checked negative result that materially redirects the search. Fragment results ($r^{\{n\}}$-free) and constant-relaxed versions count as bridges, per Remark 1.

8 Notation–Isabelle dictionary

here	Isabelle name	file
$\lvert r\rvert$, $\lVert rs\rVert$	rsize, rsizes	BI
$\lVert U\rVert$ (set)	rsize_set	AFT
flts, nub, $\oplus$	rflts, rdistinct, rsimp_ALTs	BI
σ, $\sigma^\star$	rsimp4_SEQ_atom, rsimp7_SEQ_atom	BI
pder_c, npder_c	rpder_list, rpder_norm_list	GRB
step_c, F	afactored_step, afactored1	AFT
P, prune	rsimpStrong_prune_pair_raw, ..._prune_rows_raw	GRB
S	rsimpStrong_raw	GRB
Δ_c	rpder_strong_rows_raw	GRB
D	row_dlforms, row_dlformss	AFT
D^{list}, listcost	row_dlforms_list, afactored1_strong_dlform_list_cost	AFT
fronts, pairbudget	strong_derivative_front_terms, raw_shared_prune_active_suffix_pair_budget	AFT / GRB
target gate	assumption actual_cubic of ..._tight_cubic_contractl	AFT

AFT = `AntimirovFactoredTransition.thy`; GRB = `GeneralRegexBound.thy`; BI = `BasicIdentities.thy`.

Live status, checked-facts table and dead-route list: `MAINLINE.md`; document catalog: `DOC_INDEX.md`; day-by-day state: tail of `PROGRESS_BACKREF.md` (all at the repository root).